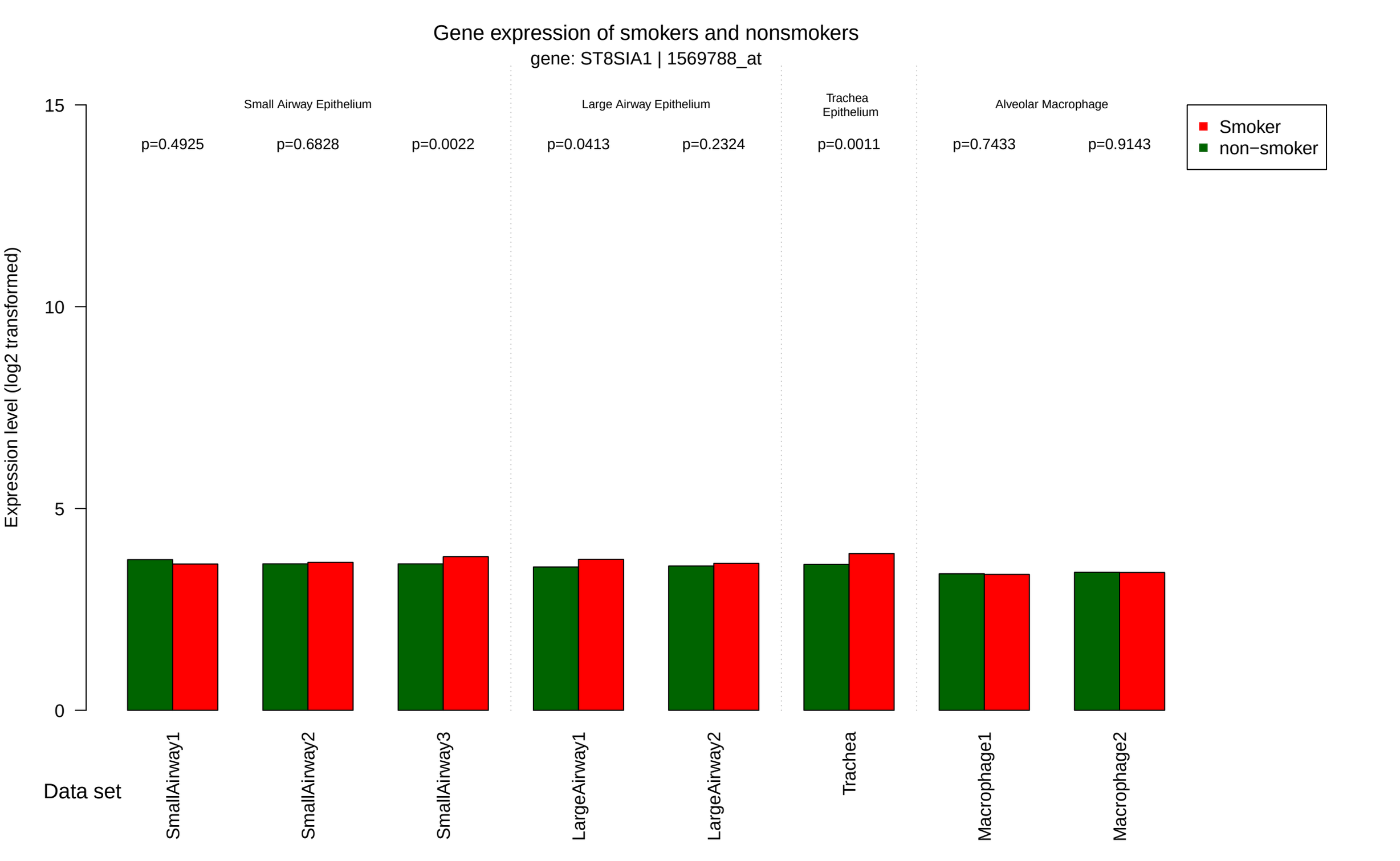
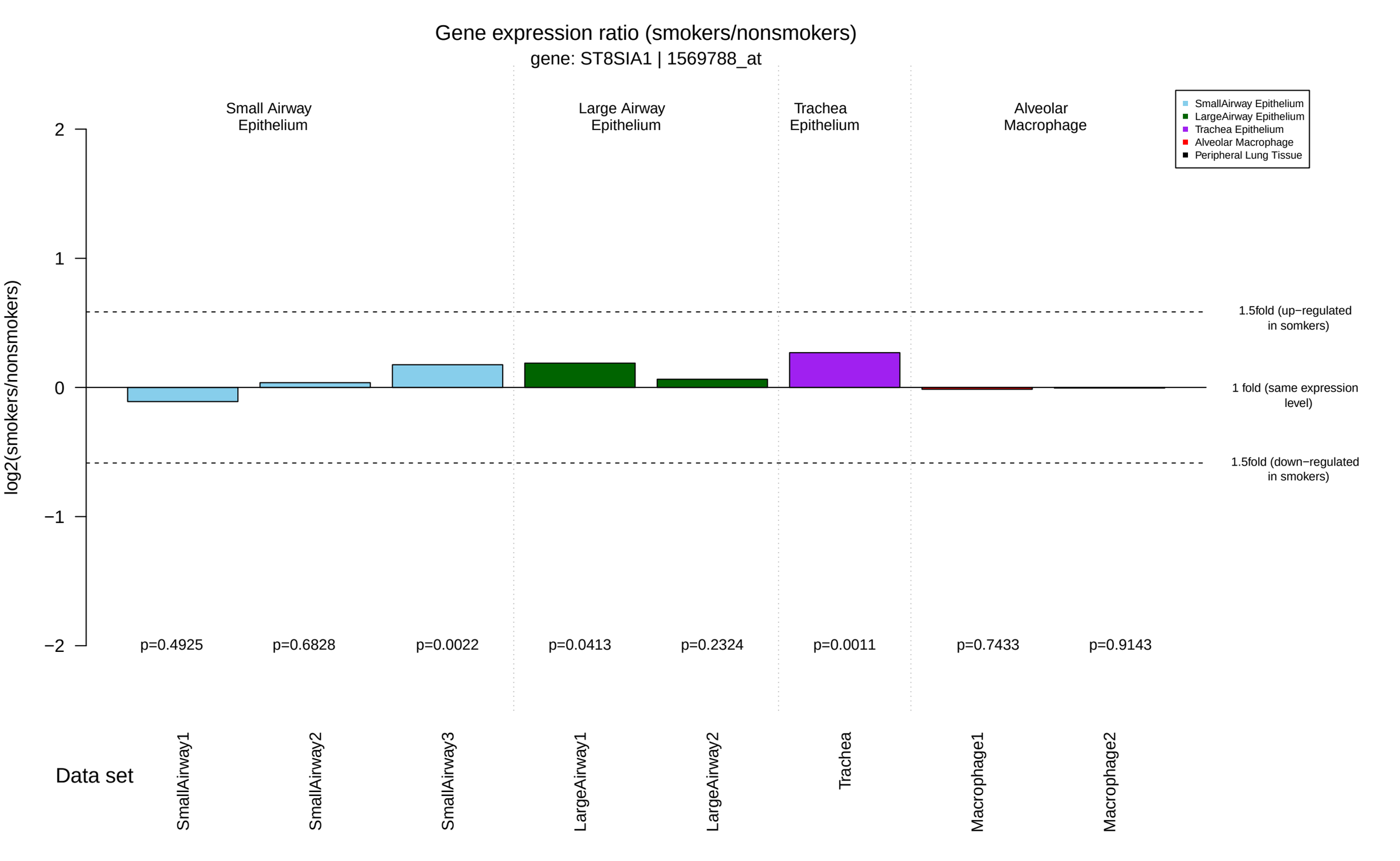
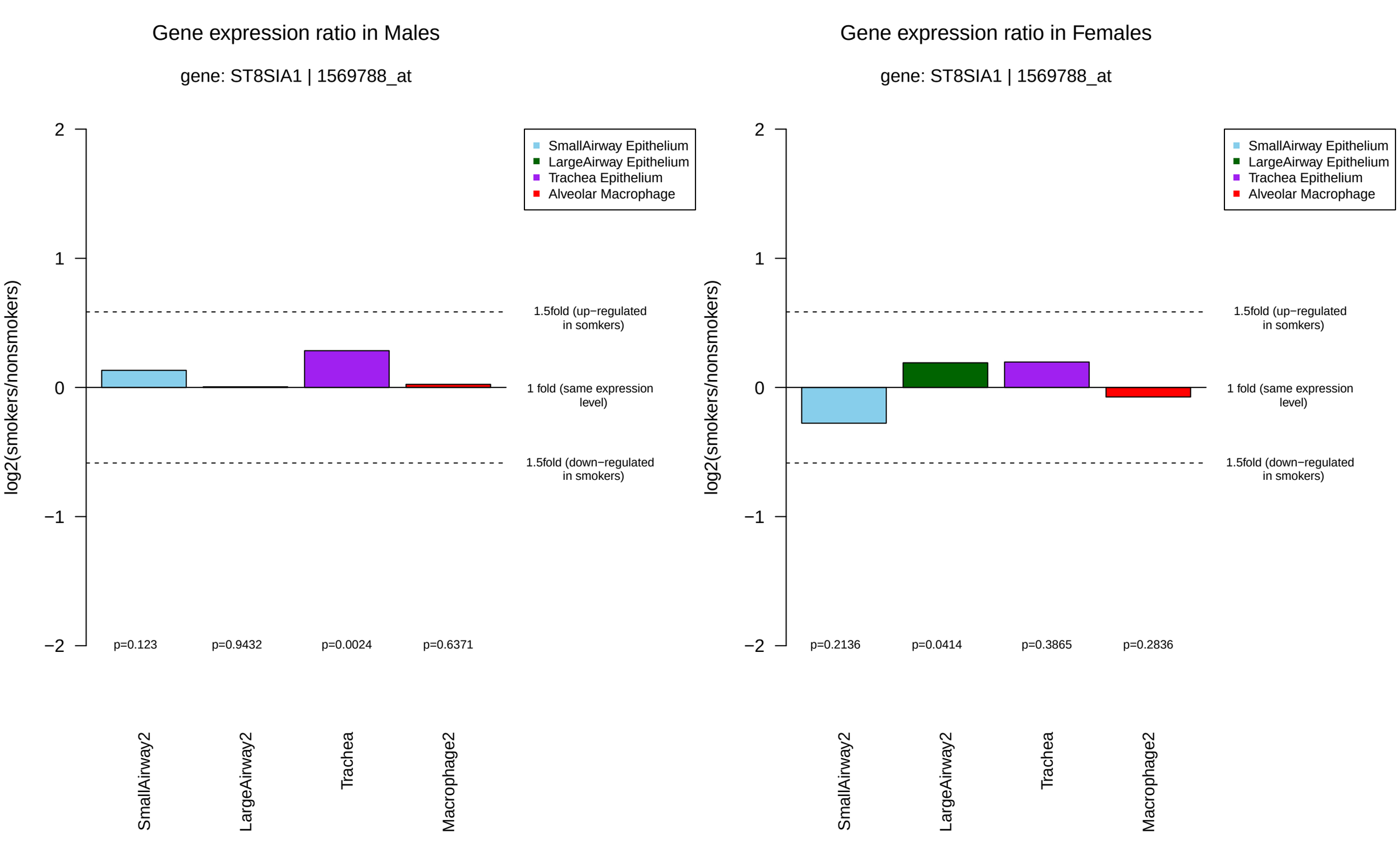
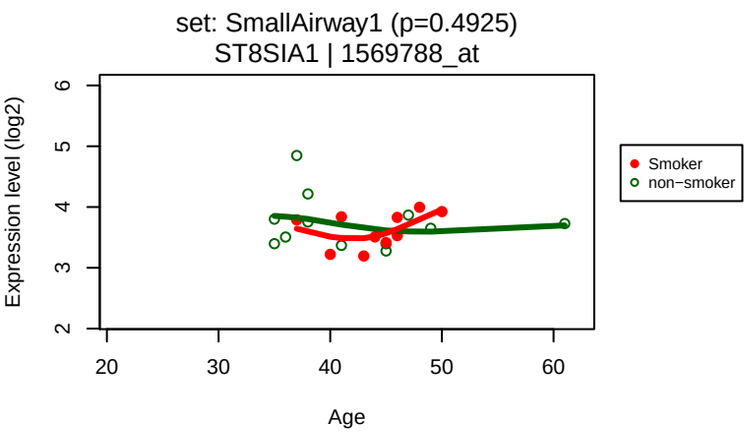
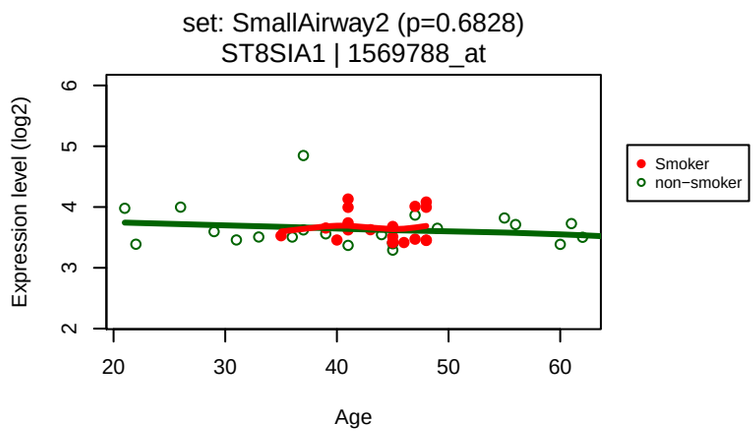
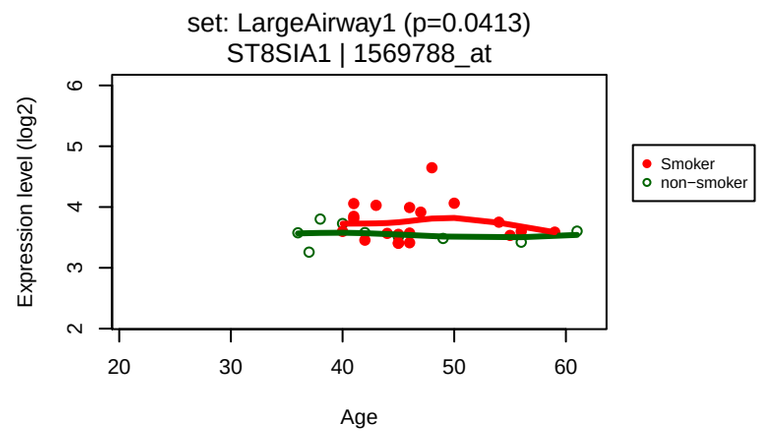

Gene expression of smokers and nonsmokers
gene: ST8SIA1 | 1569788_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4925
p=0.6828
p=0.0022
p=0.0413
p=0.2324
p=0.0011
p=0.7433
p=0.9143
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

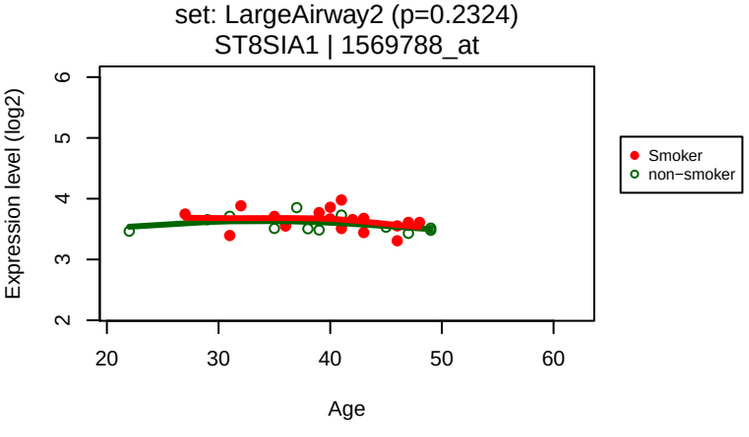

Gene expression ratio (smokers/nonsmokers)
gene: ST8SIA1 | 1569788_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4925
p=0.6828
p=0.0022
p=0.0413
p=0.2324
p=0.0011
p=0.7433
p=0.9143
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

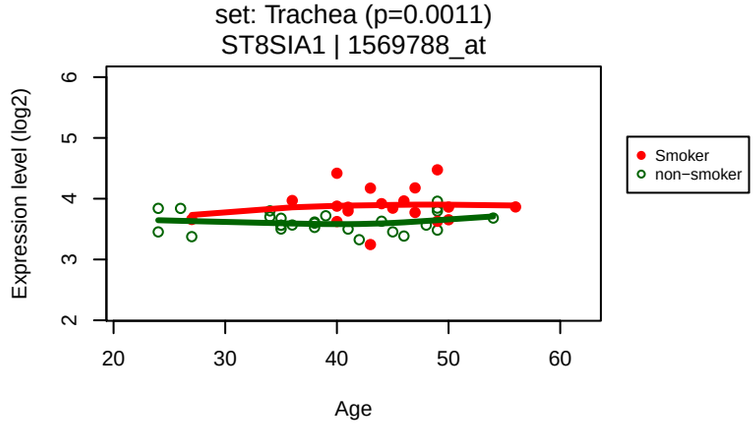

Gene expression ratio in Males
gene: ST8SIA1 | 1569788_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.123
p=0.9432
p=0.0024
p=0.6371
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ST8SIA1 | 1569788_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2136
p=0.0414
p=0.3865
p=0.2836
SmallAirway2
LargeAirway2
Trachea
Macrophage2

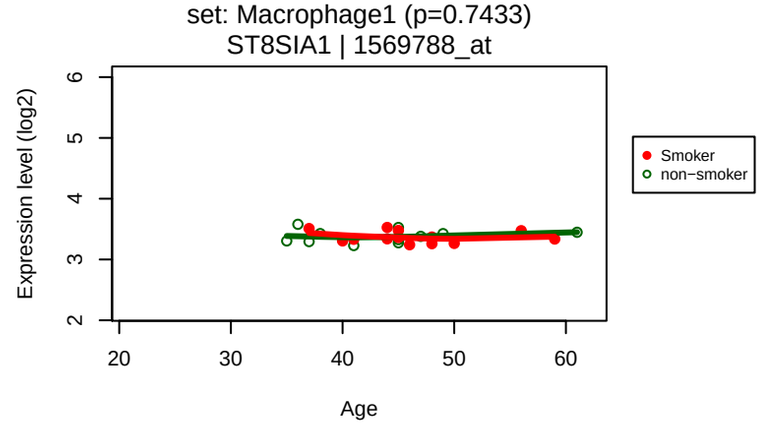

set: SmallAirway1 (p=0.4925)
ST8SIA1 | 1569788_at
Expression level (log2)
Age
Smoker
non−smoker

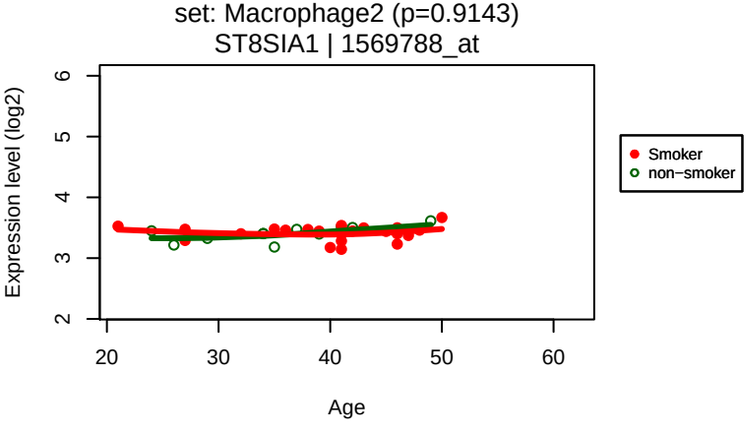

set: SmallAirway2 (p=0.6828)
ST8SIA1 | 1569788_at
Expression level (log2)
Age
Smoker
non−smoker

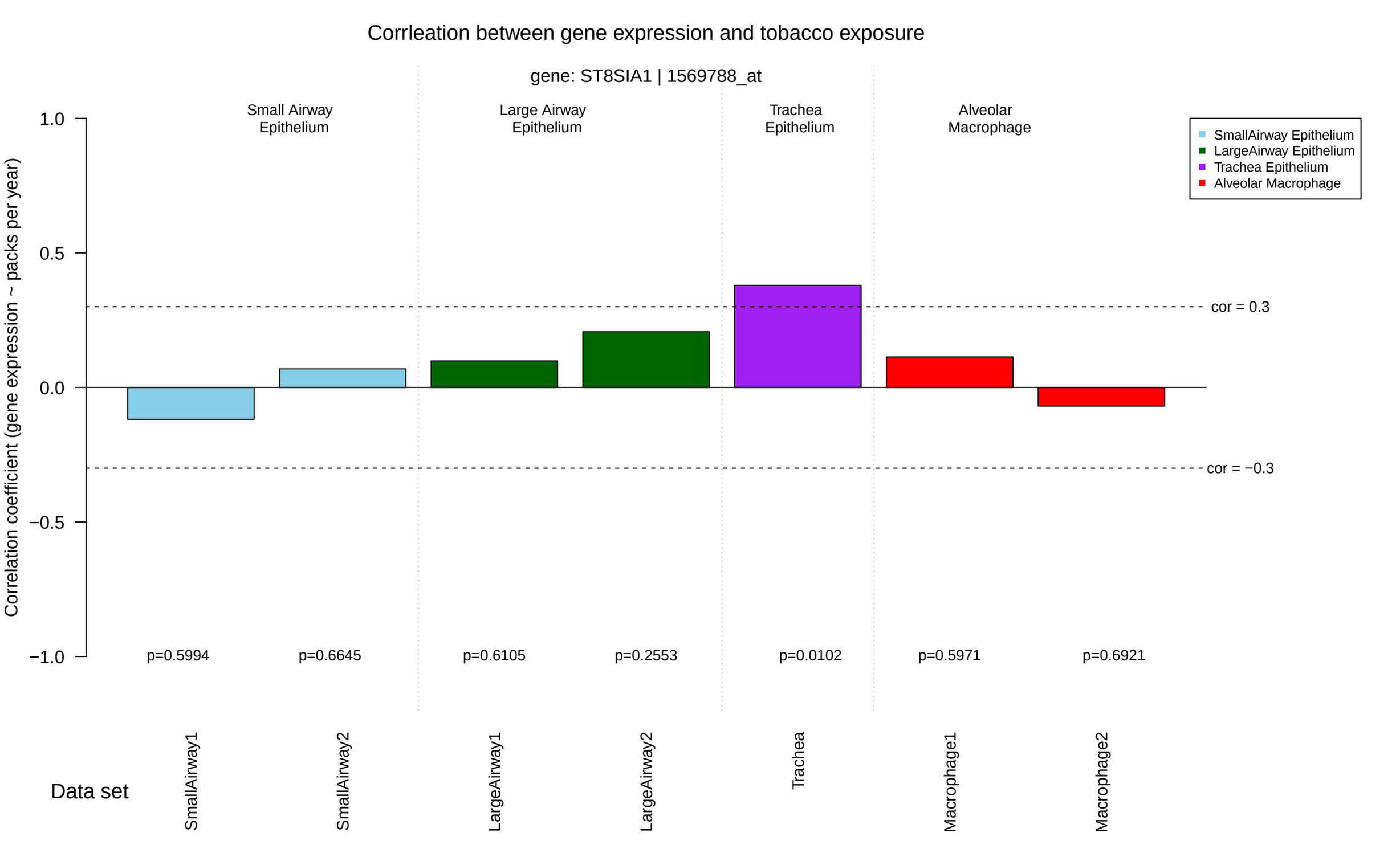

set: LargeAirway1 (p=0.0413)
ST8SIA1 | 1569788_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2324)
ST8SIA1 | 1569788_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0011)
ST8SIA1 | 1569788_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7433)
ST8SIA1 | 1569788_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9143)
ST8SIA1 | 1569788_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ST8SIA1 | 1569788_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5994
p=0.6645
p=0.6105
p=0.2553
p=0.0102
p=0.5971
p=0.6921
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2